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The Walking Corsi Test (WalCT): A Normative Study of Topographical Working Memory in a Sample of 4- to 11-Year-Olds

**L. Piccardi^{1,2}, L. Palermo^{2,3}, M. Leonzi¹, M. Risetti⁴,
L. Zompanti², S. D'Amico¹, and C. Guariglia^{2,3}**

¹Life, Health and Environmental Science Department, University of L'Aquila, Coppito, Italy

²Neuropsychology Unit, I.R.C.C.S. Fondazione Santa Lucia, Rome, Italy

³Psychology Department, University Sapienza of Rome, Italy

⁴NEILab-Neurofisiopatologia, I.R.C.C.S. Fondazione Santa Lucia Rome, Italy

We report normative data on topographical working memory collected through the Walking Corsi Test (WalCT; Piccardi et al., 2008) for developing a standard administration procedure to be used in clinical and educational practice. A total of 268 typically developing Italian children aged 4–11 years performed both WalCT and Corsi Block-Tapping Test (CBT; Corsi, 1972) a well-known visuo-spatial memory test. WalCT has already been validated in adults, demonstrating sensitivity in detecting topographical memory deficits even in individuals who have no other memory impairments. Our results showed that age, but not sex, affected performances. Both girls and boys had a larger span on the CBT than the WalCT. The youngest group did not differ in performing WalCT and CBT, but from 5.6 years of age children performed better on CBT than WalCT, suggesting that memory in reaching space develops before topographical memory. Only after 5 years of age do children learn to process specifically topographical stimuli, suggesting that this happens when their environmental knowledge becomes operational and they increase environmental independence. We also discuss the importance to introduce WalCT in the clinical assessment.

Keywords: Pathway memory; Visuo-spatial memory; Working memory; Topographical orientation; Human navigation; Corsi Block-Tapping Test.

INTRODUCTION

Animal models of spatial cognition (e.g., Rizzolatti, Berti & Gallese, 2000) show that the space around us is a multifactorial construct of our brain and that distinct areas are responsible for coding space outside reaching distance (far or extrapersonal space), within reaching distance (near or peripersonal space), and body surface space (personal or bodily space). Our brain distinguishes between far and near space; indeed, the relevance of actions performed by others may be different for the observer depending on the regions of space in which they are executed and may lead to different behavioral responses.

Gabbard, Cordova, and Ammar (2007) compared typically developing children 5–11 years of age and adults in estimating the reachability of targets presented

Address correspondence to: Dr. Laura Piccardi, Ph.D., Life, Health and Environmental Science Department, Università degli Studi di L'Aquila, P.le S. Tommasi, 1, 67100 L'Aquila, Italy.
E-mail: laura.piccardi@cc.univaq.it

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randomly at seven midline locations. They found that children made more errors in estimating reaching actions in extrapersonal than peripersonal space, whereas adults made the same number of errors in the two spaces. Gabbard and co-workers' study suggests that children have more difficulty estimating reachability outside their peripersonal space. It is reasonable to hypothesize that near and far space develop at different ages; for example, it is more important for toddlers to act in near space. In support of this view is evidence of different types of visuo-spatial memory devoted to short- and long-term storage as well as to recalling elements in near and far space (i.e., object-location vs. environmental navigation). For example, Piccardi et al. (2010, 2013) found that patients with pervasive topographical disorientation failed on memory tasks performed in far space but not in near space. Furthermore, Piccardi et al. (2008) created a large-scale version of the well-known Corsi Block-Tapping Test (CBT; Corsi, 1972), called the Walking Corsi Test (WalCT), in which the platform of the CBT is reproduced on the floor (10:1 scale). To perform the test participants have to follow the same path previously walked by the examiner. The authors developed this large-scale version to measure topographical memory, because they suspected that the CBT might not detect navigational deficits (see Piccardi et al., 2010, 2013; see also Bianchini et al., 2010). They also found gender differences in college students and age-related differences in a sample sub-divided into two groups: young (mean age 23.1; *SD* 2.01 years) and senior (mean age 62.68; *SD* 6.42 years) in performance on the two tests (Piccardi et al., 2008, 2011, 2013). In particular, concerning gender differences the authors reported that on both tests men had a larger span than women. Further, men showed a larger span on the WalCT than on the CBT, while women's spans did not differ on the WalCT and the CBT (Piccardi et al., 2008). In respect to age-related differences young participants had a larger span than senior participants on both tests (Piccardi et al., 2011), and further, WM performance on WalCT started to decrease at 47 years of age (Piccardi et al., 2013). The presence of gender and ageing differences in performing the two tests further supports the idea that visuo-spatial working memory (VSWM) may be dissociated in peripersonal and extrapersonal space.

VSWM is usually framed within some theoretical models. According to Baddeley's model VSWM is considered a multiple-component cognitive system, comprised of a central executive and two slave sub-systems: the phonological loop and the visuo-spatial sketchpad (Baddeley & Hitch, 1974; for review see Logie, 1996, 2003). The phonological loop is involved in processing and holding verbal information, while the visuo-spatial sketchpad is needed in processing and for holding visuo-spatial information. In Logie's multicomponent model (1995, 2003), the visuo-spatial sketchpad is divided into two major components: one for processing visual information (*Visual Cache*) and one for processing spatial information (*Inner Scribe*). The former provides a temporary store for visual information (i.e., color, shape, and static visual patterns) and the latter handles information about movement sequences (i.e., sequential locations or movements) and provides a mechanism through which visual information can be rehearsed in WM for implementing planning and execution of movement. The present study investigated working memory in extrapersonal (topographical) space as measured by the WalCT and in light of different components of Logie's (1995, 2003) multicomponent model. We decided to frame topographical WM in Logie's model, because in Baddeley's model (2000) WM is described as a non-unitary function (but the distinction concerns only verbal and visuospatial material). By contrast, Logie's model considers

WM a store that rehearses movement sequences. Pickering and co-workers (Pickering, Gathercole, Hall, & Lloyd, 2001) reported dissociations between visual and spatial components that seemed to develop at different rates and with little overlap in a given age group of children. Recent research has also demonstrated the possibility of fragmenting the spatial component of VSWM even more. For example, Pazzaglia and Cornoldi (1999) showed that manipulation of the sequential/simultaneous presentation of to-be-memorized VSWM stimuli is an important variable, suggesting a differentiation between sequential and simultaneous processes in the inner scribe. They suggested that the visual cache holds visual information such as form, color, texture, and static layout of objects, and the inner scribe holds both sequential and simultaneous spatial information. From a navigational point of view, distinctions within the VSWM theoretical model are extremely interesting. Indeed, we can hypothesize that sequential information is used more during navigation than simultaneous information and that topographical stimuli might be processed by the inner scribe.

The importance of developing standardized tools to assess topographical memory is suggested by the recent description of developmental topographical disorientation (DTD; Iaria, Bogod, Fox, & Barton, 2009; see also Bianchini et al., 2010, 2013). DTD is a complex syndrome involving different aspects of topographical skills with different degrees of severity among cases, that affects individuals without brain lesions, or neurological or psychiatric disorders, and with normal development of other cognitive functions. Since a feature characterizing a kind of DTD is the presence of topographical memory deficits (Bianchini et al., 2010) the WalCT can help clinicians in early diagnosis of DTD that is not so rare a disorder as hypothesized at the beginning. Indeed, a recent study by Iaria and Barton (2010) describes 120 new cases of individuals affected by DTD from birth, suggesting that DTD may be more diffuse than expected. Their findings strongly suggest that DTD may be a new developmental learning disability with a strong impact on society. If we consider that navigation has a crucial role in the adaptive daily life behavior, and that it could be impaired in adolescents who were born premature and suffer congenital bilateral periventricular brain damage (Pavlova, Sokolov, & Krägeloh-Mann, 2007), we can understand the importance of having tools to assess navigation in children. These tools might help in testing navigational ability in a systematic way for then developing suitable rehabilitative programs when necessary.

In the present study we aimed to investigate the psychometric properties (i.e., correlation with other memory tests, namely the CBT) of the WalCT in a group of 268 typically developing Italian children to create a standard administration procedure with normative data of a working memory task for children between the ages of 4 and 11 years for use in clinical practice. We also interpret our data by discussing how topographical working memory might be inserted in Logie's WM theoretical frame (Logie, 1995, 2003).

METHOD

Participants

The present study was carried out in 268 typically developing Italian children (132 M and 136 F) who had no reported school difficulties. Ages ranged from 4 years

to 11 years and 1 month (mean age = 87.75 months; *SD* = 19.90 months). Participants were subdivided into six groups according to age:

- 37 children (C1: 16 males, MC1, and 21 females, FC1) aged from 4.0 to 5.3 years (48–64 months)
- 50 (C2: 29 males, MC2, and 21 females, FC2) aged from 5.4 to 5.9 years (65–74 months)
- 50 (C3: 22 males, MC3, and 28 females, FC3) aged from 6.0 to 6.7 years (75–81 months)
- 51 (C4: 25 males, MC4, and 26 females, FC4) aged from 6.8 to 7.9 years (82–95 months)
- 42 (C5: 23 males, MC5, and 19 females, FC5) aged from 8.0 to 9.2 years (96–111 months)
- 38 (C6: 17 males, MC6, and 21 females, FC6) aged from 9.3 to 11.1 years (112–133 months).

None had visual or hearing impairments or had been diagnosed with a neurological condition. Foreign children and those with learning difficulties (as reported by their teachers or families) were not included in the study.

The study was approved by the local ethical committee in accordance with the Declaration of Helsinki. A signed consent form was obtained from parents and an assent from each child.

Children were tested individually in a quiet room of the primary school. Data were collected in the following primary schools: “Suore della Presentazione” in Pescara; “Istituto Suore Dottrina Cristiana” in L’Aquila; and “Istituto Giovanni Pascoli” in Aprilia (LT).

Tests and procedure

Corsi Block-Tapping Test (CBT; Corsi, 1972). The CBT consists of nine wooden blocks (4.5×4.5 cm) fixed on a baseboard (30×25 cm) in a scattered array. On the experimenter’s side the blocks are numbered for easy identification. The experimenter taps a number of blocks at a rate of one block per 2 seconds using the index finger for tapping the blocks and lifting the hand straight up (for 2 seconds) before moving it to the next block. After this the participant has to tap the block sequence in the same order it was presented. The block sequences gradually increase in length (i.e., number of blocks starting from a two-block sequence), and the score is the number of blocks in the longest sequence remembered correctly (block span). Five sequences are presented for every length: if the participant correctly reproduces at least three out of five sequences the test continues with the next length; otherwise it is stopped.

Participants were tested individually in a quiet room of the primary school that had artificial lighting. They were seated facing the examiner on a height-adjustable office chair in front of the CBT baseboard.

The Walking Corsi Test (WalCT: Piccardi et al., 2008). A larger version of the CBT (3×2.5 m; scale 1:10 of the CBT) was set up in an empty room of the primary school. In this test the participant has to actually walk and reach different

locations. The WalCT consists of nine black squares placed on an unvarying floor in the same position as in the standard CBT (see Figure 1).

Although the same experimental conditions were adopted in the WalCT and the CBT, the starting position was different. In the WalCT, but not in the CBT, both the examiner and the participants started from the same point. The location of the starting position was decided in a pilot study in which different points of view did not influence performance on the WalCT (as reported in Piccardi et al., 2008). This was done to facilitate replication of the test in rooms that are slightly larger than the layout of the test. The experimenter illustrated the sequence by walking on the layout and stopping on each square for 2 seconds. Then the participant had to repeat the same sequence as the experimenter by walking and stopping on the squares included in the sequence. In the WalCT, participants obtained a square span corresponding to the longest sequence remembered correctly. Sequences administered in the WalCT are reported in Appendix 1.

The administration order of the CBT and WalCT was counterbalanced across the participants in each group, and participants performed the two tests in two separate sessions on 2 days in a row, to avoid a practice effect that would potentially give an advantage to the second test performed.

RESULTS

Table 1 and Figure 2 show means and standard deviations (*SD*) for female and male participants in each group in both the CBT and the WalCT.

In order to rule out an effect related to the administration order of CBT and WalCT we counterbalanced presentation order across the participants in each group. Then we performed a 2 (type of test: CBT vs. WalCT) $\times$ 2 (administration order: CBT-WalCT vs. WalCT-CBT) mixed ANOVA on span scores. The analysis revealed a main effect of the test (CBT: $M = 3.63$, $SD = 0.97$; WalCT: $M = 2.58$, $SD = 1.13$); $F(1, 266) = 284.98$; $p < .001$, but not for the administration order, $F(1, 266) = .004$; *ns*, or for the interaction between type of test and administration order, $F(1, 266) = 0.13$; *ns*.



Figure 1. Child performing WalCT. The scale was 10:1 of CBT and the apparatus measured 2.5 m $\times$ 3m; black squares were 30 cm $\times$ 30 cm.

Table 1. Means (*SD*) in Corsi Block-Tapping Test (CBT) and Walking Corsi Test (WalCT)

Group	CBT (<i>n</i> of cubes)	WalCT (<i>n</i> of squares)	<i>N</i>
FC1 (range 48–64 months)	2.62 (1.12)	1.90 (1.04)	21
FC2 (range 65–74 months)	2.62 (1.00)	1.90 (1.18)	21
FC3 (range 75–81 months)	3.54 (0.79)	2.43 (0.84)	28
FC4 (range 82–96 months)	3.81 (0.57)	3.08 (0.98)	26
FC5 (range 97–111 months)	4 (0.75)	3.63 (0.60)	19
FC6 (range 112–133 months)	4.57 (0.75)	3.09 (1.3)	21
MC1 (range 48–64 months)	2.63 (0.81)	1.94 (1.12)	16
MC2 (range 65–74 months)	3.69 (0.66)	2.34 (1.14)	29
MC3 (range 75–81 months)	3.73 (0.63)	2.14 (1.04)	22
MC4 (range 82–96 months)	3.4 (0.76)	2.48 (1.19)	25
MC5 (range 97–111 months)	4.04 (0.88)	2.87 (0.76)	23
MC6 (range 112–133 months)	4.53 (0.72)	3.06 (0.90)	17

In table means and standard deviations (*SD*) are reported for both CBT and WalCT spans (number of cubes/squares). Groups are subdivided both for age and sex, the number of children for each group is listed in the last column. F = female participants; M = male participants; *N* = number of participants.

Pearson correlations showed that the CBT and the WalCT were significantly correlated with each other ($r = .53$; $p < .001$). Pearson correlations were also calculated to determine whether performance on the CBT and the WalCT was correlated with age, and Point Biserial correlations were calculated to determine whether performance on the CBT and the WalCT was correlated with gender. We found that age was significantly correlated with both the CBT ($r = .57$; $p < .001$) and the WalCT ($r = .4$; $p < .001$). Sex, however, was not correlated with either the CBT, $r_{pb} = +0.06$; $t(1, 268) = 0.91$, *ns*, or the WalCT, $r_{pb} = -0.09$; $t(1, 268) = -1.39$, *ns*.

Standard mixed linear regression analyses were carried out to investigate whether age (expressed in months) as continuous variable, sex, and tests order of administration as categorical predictor variables predicted performance on both the CBT and the WalCT. The regression with CBT as dependent variable and age, sex, and tests order as independent variables showed that only age ($\beta = 0.564$); $t(264) = 11.11$, $p < .001$, predicted test performance—sex: $\beta = -0.03$; $t(264) = -0.66$, $p = .51$; tests order: $\beta = 0.01$; $t(264) = 0.14$, $p = .89$). Similarly, the regression analysis with the WalCT as dependent variable showed that only age ($\beta = 0.41$); $t(264) = 7.24$, $p < .001$, predicted performance on the test—sex: $\beta = 0.10$; $t(264) = 1.81$, $p = .07$; tests order: $\beta = -0.01$; $t(264) = -0.21$, $p = .83$) (Figure 3).

($p < .001$), and C6 ($p < .001$) groups, and the C6 group performed best because their scores were significantly higher than the C1 ($p < .001$), C2 ($p < .001$), C3 ($p < .001$), and C4 ($p < .05$) groups. Moreover, the C2 and C3 groups performed worse than the C5 (respectively $p < .001$ and $p < .005$) group. No differences were found in the performances of the other groups.

A significant sex $\times$ tests, $F(1, 256) = 5.96$; $p < .01$, sex $\times$ group, $F(5, 256) = 3$; $p = .01$, and group $\times$ test, $F(5, 256) = 4.56$; $p < .001$, interaction was found, but not a significant group $\times$ test $\times$ sex interaction, $F(5, 256) = 0.39$; *ns*.

A post hoc analysis (Scheffé Test) carried out to investigate the sex $\times$ test interaction showed that both females and males had a larger span on the CBT than the WalCT ($p < .001$). No gender differences were found on either the CBT or the WalCT.

The post-hoc analysis carried out to investigate the sex $\times$ group interaction showed that the FC1, FC2, and MC1 performed worse because they scored significantly lower than the FC4 (respectively $p < .01$; $p < .05$; $p < .05$), FC5 ($p < .001$), FC6 ($p < .001$), MC5 (respectively $p < .01$; $p < .05$; $p < .05$) and MC6 ($p = .001$), who performed best. No differences were detected among the other groups.

The Scheffé Test performed to investigate the group $\times$ test interaction showed that performance of both C1 and C2 on the WalCT was significantly lower than that of C4 ($p < .001$ and $p < .05$ respectively), C5 ($p < .001$), and C6 ($p < .001$), and that C3 performed worse than C5 ($p < .001$) and C6 ($p < .01$) on the WalCT. No differences were found in the other groups on the WalCT.

On the CBT, C1 scored significantly worse than C2 ($p = .05$), C3 ($p < .001$), C4 ($p < .001$), C5 ($p < .001$), and C6 ($p < .001$), and C2 scored worse than C5 ($p = .01$) and C6 ($p < .01$). The C6 group performed best because scores were significantly higher than in the C1 ($p < .001$), C2 ($p < .001$), C3 ($p < .001$), and C4 ($p < .001$) groups. No differences were detected among the other groups on the CBT.

Furthermore, the C2, C3, C4, C5 and C6 groups performed better on the CBT than the WalCT ($p < .01$). Instead no differences were detected in performance on the WalCT and the CBT in the youngest (C1) group.

DISCUSSION

Our data suggest there is a developmental trend characterized by slow but constant improvement in performance over time on both the CBT and the WalCT. We observed that the CBT span is larger than the WalCT span, which supports the idea that visuo-spatial competence develops first in peripersonal space and subsequently in extrapersonal space. One interesting finding is that the WalCT and CBT spans did not differ in the 5-year-olds, showing that children learn to process topographical stimuli differently only when their environmental knowledge becomes operational and they have increased environmental independence. At that point, the ability to choose and use navigational problem-solving strategies increases until it becomes automatic due to repeated use. Generally, topographical span is smaller than spatial span. From 6 years of age a range of advantage of about 0.37–1.48 spatial span items over the topographical span can be observed, and for both of them slow but constant improvement in performance can be observed over time. We did, however, observe an important difference in CBT and WalCT spans between 8- to 10-year-olds and 6- to 7-year-olds. We found no gender differences in topographical and spatial WM. This result

concerning spatial WM differed from that reported by Orsini et al. (1987; Orsini, Schiappa, & Grossi, 1981) that found sex differences both in adults and children (age range: 4–10 years) but it is in line with Capitani, Laiacona, and Ciceri (1991), who never found a gender effect in adults. Concerning sex differences in age of development, results are largely conflicting. Smirni, Villardita, and Zappalà (1983) assessed spatial memory span in 195 young adolescents using the CBT and found no sex differences. Similar results were also reported by Villardita, Smirna, le Pira, and Zappalà (1981), who observed no significant differences in verbal, visual, and spatial (with meaningful and meaningless figures) learning tasks in 122 adolescents, although they found that males performed better than females in the visual serial learning test with meaningless figures. Farrell Pagulayan, Busch, Medina, Bartok, and Krikorian (2006) did not find sex differences in the spatial span capacity of 340 participants aged 6 to 13 years, and Nichelli, Bulgheroni, and Riva (2001) observed no differences in performance between boys and girls aged 5 to 14 years. In any case, the large variability in results might just reflect differences in sample size, administration, and scoring parameters across studies. For instance, as suggested by Berch, Krikorian and Huha (1998), few investigators reported the manner in which the blocks were pointed to by the examiner, but this could be one of the parameters that might account for the presence of individual differences. Indeed, Nichelli et al. (2001), the study that did not find gender differences, used the procedure in which, after pointing to each cube, the examiner returned the forefinger to the surface of the table, while Orsini et al. (1987), the study that found gender differences, did not indicate their procedure. Further, Orsini et al. (1987) had the larger sample of children assessed in performing CBT (1112 children) than Nichelli et al. (2001) that tested 275 children and our study in which we assessed 268 children. These discrepancies among studies might have arisen from differences in sample size and or sample selection.

A strong relationship between working memory and navigation has been confirmed. For instance, Smith, Gilchrist, and Hood (2005) demonstrated that children with a longer Corsi span completed the trials of a searching task in a large-scale space in less time and with fewer visits than children with a shorter span. Topographical working memory per se has never been considered. The present results point out that different types of visuo-spatial WM are present in infancy and develop along different lines. The evidence that WM for peripersonal space develops earlier than WM for topographical space demonstrates the greater effort needed to reproduce a sequence of steps in a large-scale context than to reproduce a sequence of reaching movements in a small-scale context. It is possible that differences among tests' performances could be related to the inter-stimulus time and the differences among working memory demands. On the CBT the child can respond quickly by pointing rapidly, minimizing the working memory demands of the task. On the WalCT the speed with which the child can respond is limited by the distance between squares. However, an interesting study by Beigneux, Plaie, and Isingrini (2007)—in which the effect of aging on the storage of visual and spatial working memory was measured by Visual Patterns Test and CBT—also showed that when the encoding time between the Visual Patterns Test and the CBT is controlled results are comparable to those of the experiment in which encoding times were not the same. The authors suggest that age-related effect on the visual span is robust and independent of the encoding time. Further, Piccardi et al. (2008) observed in college students a larger span in WalCT than CBT despite the longer inter-stimulus rate of WalCT. In the

literature the inter-stimulus interval has little or no effect on performances in not-elderly populations (Neils, Newman, Hill, & Weiler, 1991; Skottun, 2004). Alternatively, we believe that children have a greater chance to explore and use reaching space than topographical space. Boys and girls performed the WalCT without differences, suggesting that sex differences in topographical WM appear later, as found in young adults by Piccardi et al. (2008), and emphasizing the importance of hormonal and social factors (Voyer, Voyer, & Bryden, 1995) in determining sex differences in performance.

Individuals affected by developmental topographical disorientation show specific deficits in topographical memory with respect to preserved ability in visuo-spatial memory for reaching space (Bianchini et al., 2010). Iaria et al. (2009) found a lack of activation in the hippocampal complex and retrosplenial cortex in an fMRI navigational task performed by a participant with DTD.

More recently an fMRI study by Nemmi, Boccia, Piccardi, Galati, and Guariglia (2013) confirmed the importance of retrosplenial cortex in performing WalCT. Indeed, the authors investigated the regions involved in the performance of WalCT, finding activations in right dorso-lateral pre-frontal cortex, calcarine sulcus, and lingual gyrus; specifically, the region in which the calcarine sulcus joins the parieto-occipital sulcus is part of the retrosplenial complex (Epstein, 2008; Epstein, Parker, & Feiler, 2007), a region found to be active in several navigational tasks (Iaria, Chen, Guariglia, Ptito, & Petrides, 2007; Maguire, Burgess, Donnett, & Frackowiak, 1998). These results support the importance of WalCT in early diagnoses of DTD, a disorder that was placed in the category of selective developmental disorders which also includes dyslexia, developmental prosopagnosia, and selective language delay (Bianchini et al., 2013). Furthermore, as its estimated prevalence in the general population seems to be greater than initially hypothesized (Iaria & Barton, 2010), new tools for early diagnosis are mandatory to allow for early intervention.

The dissociations in processing topographical stimuli and visuo-spatial non-topographical stimuli shown by both DTD participants and brain-damaged patients with topographical disorders (Bianchini et al., 2010; Piccardi et al., 2010, 2011, 2013) indicate the importance of having a specific memory for environments. In light of Logie's WM Model (Logie, 1995, 2003), we believe that topographical stimuli might be handled by an inner scribe and that sequential components are crucial when individuals have to process pathways.

The present study confirms the importance of using different instruments to assess different types of visuo-spatial memory, particularly for early identification of DTD in children and the successive planning of precocious educational programs aimed at improving or rehabilitating wayfinding abilities. The ability to navigate improves with practice (Maguire et al., 2000; Winocur, Moscovitch, Fogel, Rosenbaum, & Sekeres, 2005) and might be compromised in individuals with restrictions in active spatial exploration. Indeed, Pavlova et al. (2007) demonstrated that congenital periventricular brain damage involving parietal and frontal regions in the right hemisphere was associated with long-lasting impairments in visuo-spatial navigation in adolescence presumably causing disintegration of the neural network engaged in successful visual navigation. Having a tool that could be easily used in clinical testing will allow early diagnosis and training of children with specific deficits in active spatial exploration and wayfinding in both familiar and unfamiliar surroundings. Indeed, the WalCT can also be used for evaluating individuals with severe motor impairments, as demonstrated by De Nigris

et al. (2013) who adapted the WalCT for assessing navigational memory deficits in adult patients with severe motor impairment following acquired brain lesions.

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APPENDIX 1

Figure A1. (a) The nine squares placed on the WalCT platform. The squares are in the same position as in the standard CBT. Numbers were reported only on the score sheet of the examiner and they were used in helping him/her to individuate sequences, as in a map. (b) The tapping sequences with the sequence lengths shown in brackets.

